MODELING OF EARTH SYSTEM AND SUSTAINABILITY · 2026

Cutting carbon in concrete without losing durability

Optimising mixes with 11 years of natural carbonation data

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Roadmap: Problem → Carbonation → Approach → Results → Drivers → Limits



THE MATERIALS

Clinker, slag and fly ash



Clinker

Baked limestone and clay nodules.
Ground with gypsum to make cement.



Slag

By-product of iron making.
Replaces part of the clinker.



Fly ash

Fine ash from coal power plants.
Replaces part of the clinker.

Less carbon, but at what cost?

7–8%

of global CO2 is from cement

- Kiln fuel + limestone releasing CO2
- Clinker holds most carbon
- Use less clinker

Trade-off

Slag and fly ash cut carbon

- But less alkaline reserve
- Carbonation moves faster
- Loss of protection increases corrosion risk

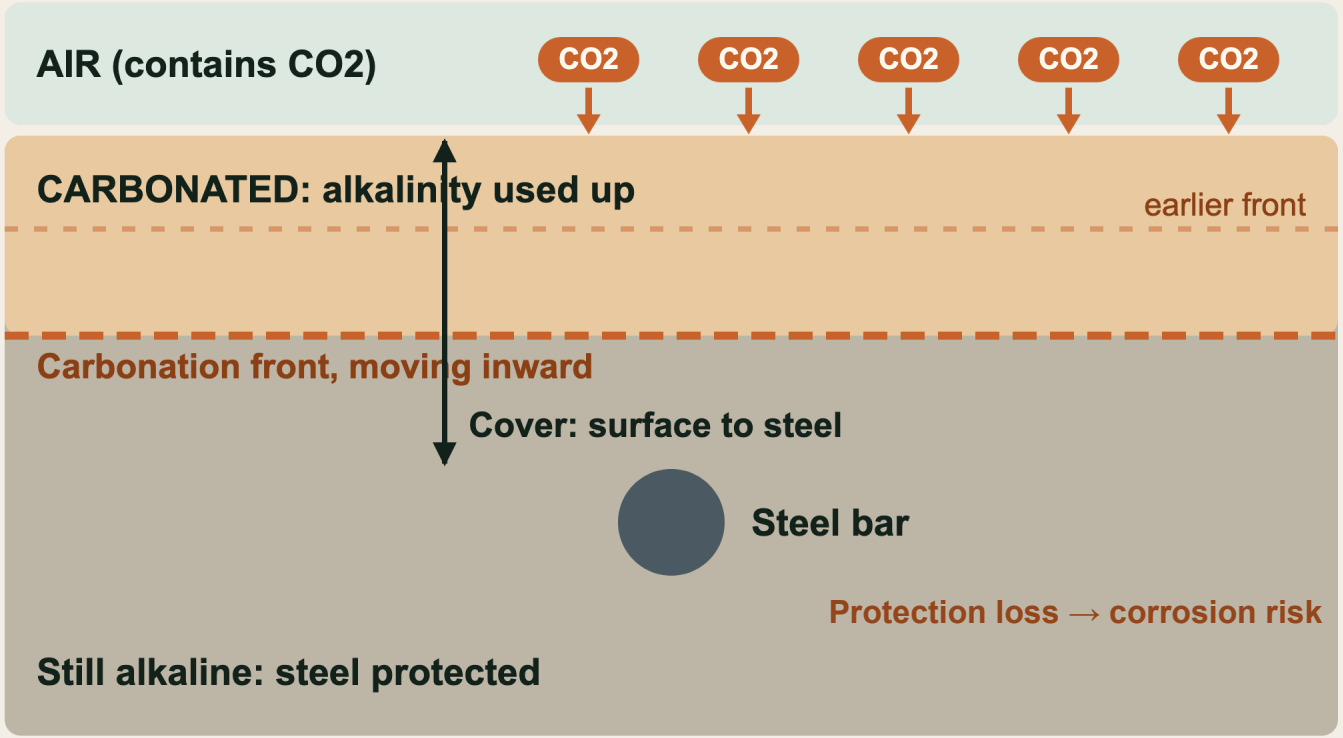
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Lowest-carbon recipe that:

- is strong enough
- protects steel for 50 years

DEFINE clinker · slag · fly ash · alkaline · carbonation · cover (surface to steel)

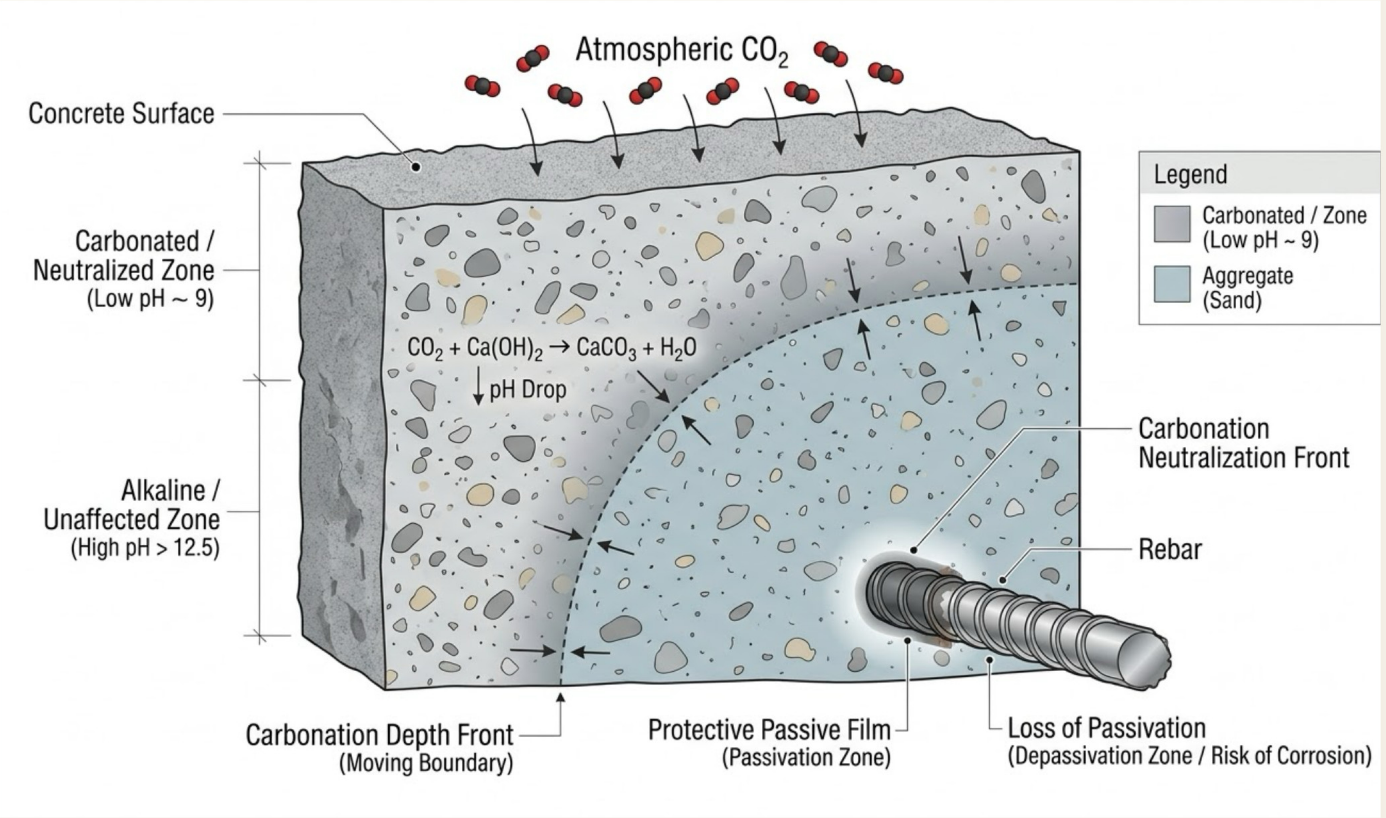
How CO2 reaches the steel



1. CO2 from the air enters at the surface
2. It uses up the alkalinity; a front moves inward
3. At steel, loss of protection increases corrosion risk

Goal: keep the front away for 50 years

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Data, models, margins, optimiser

1 Data

- 45 mixes, Chennai, 11 years
- Roof vs rain exposure
- 353 depth readings (colour spray)
- Strength and trapped air recorded

2 Models

- Strength: water ratio + air + SCM proportions; R^2 0.73
- Carbonation: grows with √time, R^2 0.87
- Uptake: CO_2 binding to calcium
- Strength model: leave one mix out at a time

3 Margins

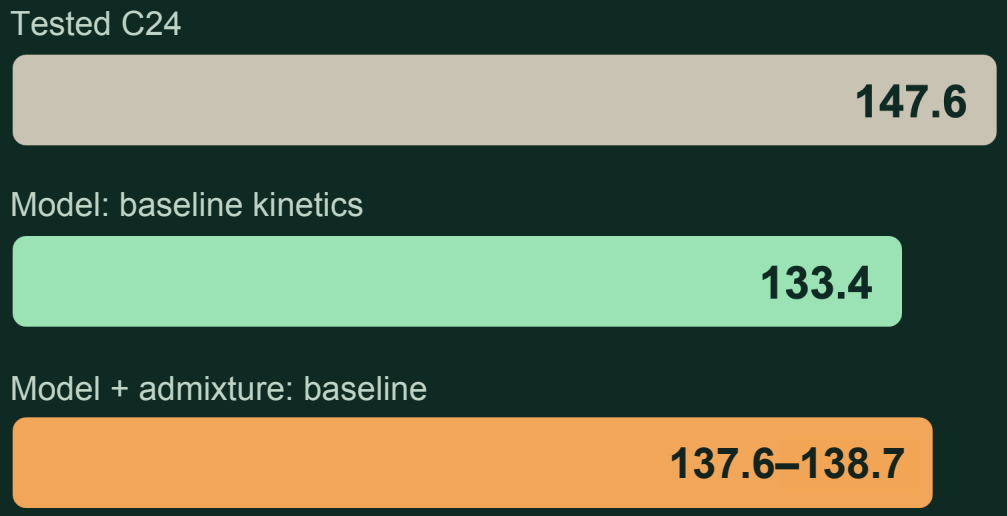
- Strength +8.25 MPa (IS 10262)
- Separate +15% kinetics sensitivity
- Minimum binder from EN 206 and IS 456
- Total SCM ≤ 50%; fly ash ≤ 35%

4 Optimise

- Searches permitted mix space
- Goal: lowest net carbon (made − reabsorbed)
- 6 variables: water ratio, binder, 4 cement replacements
- 18 cases: 3 grades × 3 covers × 2 exposures

ALSO IN THE PAPER 8 what-if checks (speed, uptake, air, code limits) · real tested mixes vs optimum · carbon value of each mm of cover

About 10% less carbon

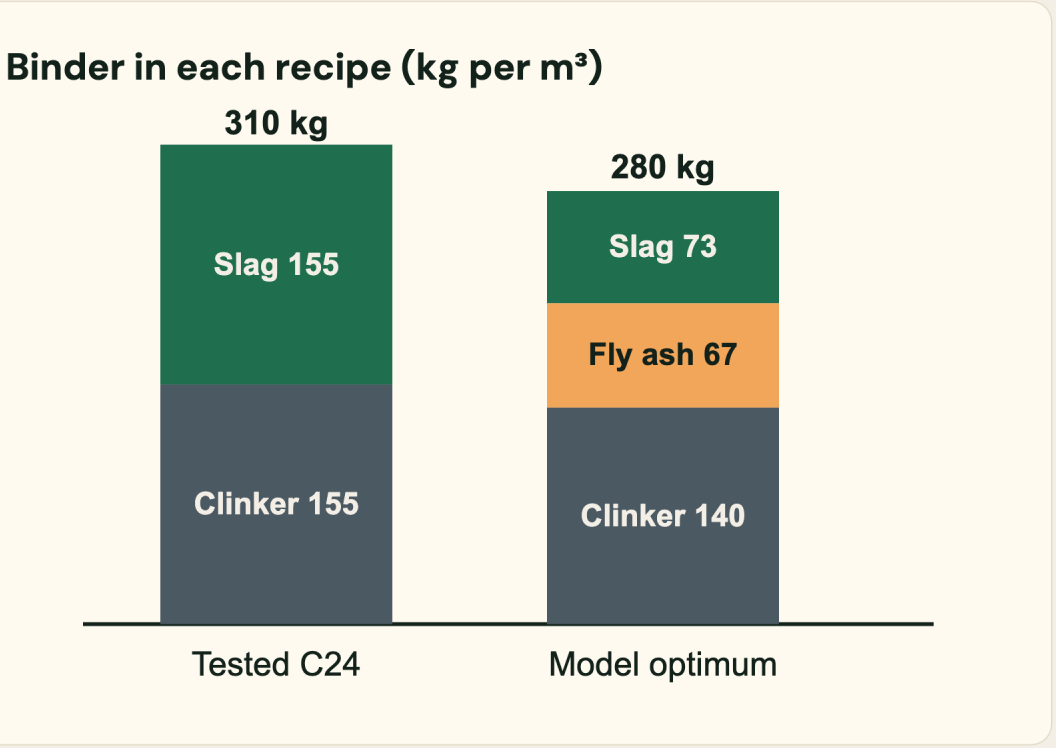


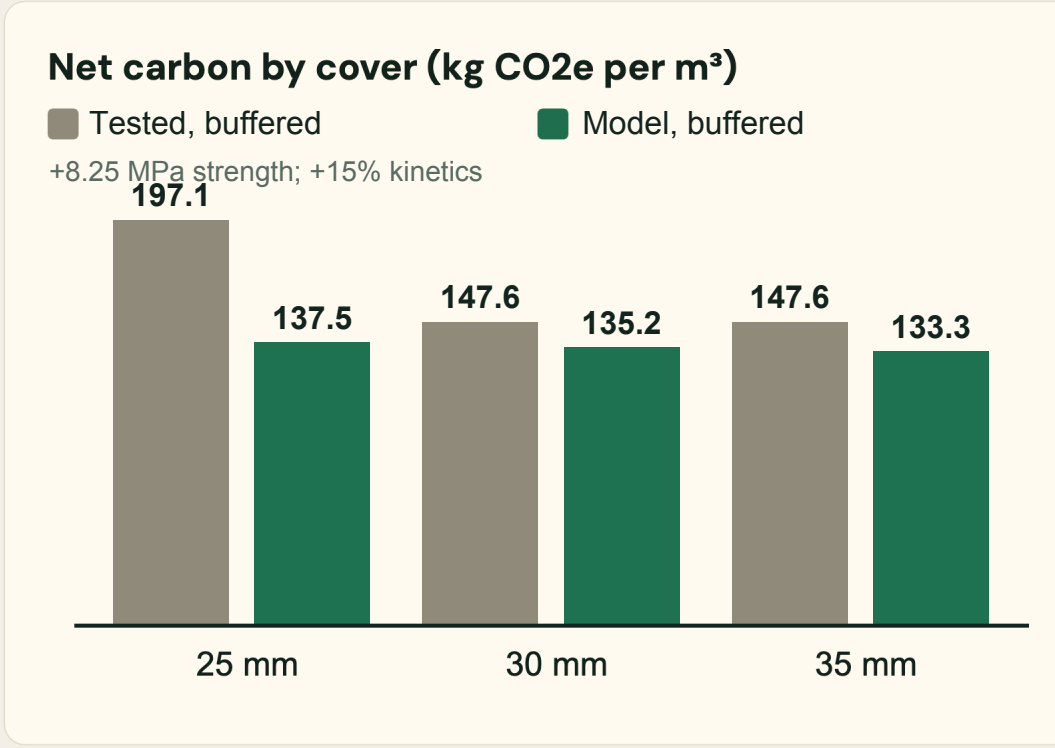
−14.1

kg CO2e/m³ saved vs tested C24 (baseline)

- Baseline kinetics: model, not tested
- w/b 0.45; 280 kg binder;
  24% fly ash + 26% slag
- Modeled PCE: 137.6–138.7 (6–7%)
- Separate +15% kinetics: 135.2
- 25 mm buffered: 197.1 → 137.5

Less clinker, and thin cover is costly





Less binder beats clever blending

79%

of saving: less binder, less water

Blending contributes 21%; binder reduction contributes most.

2.2×

fitted FA/slag effects

Per % replacement in the model. Limited by cover.

0.41

kg CO2e/m³ per extra mm cover

Local durability-controlled benchmark value.

4–6%

of emissions reabsorbed

Uptake is no fix. Cut clinker first.

A prediction, not yet a proof

Limits

- Sparse inputs at w/b 0.45; requires validation
- No tested mix used additive; 2% air assumed
- 50 years projected from 11
- Indicator front may lag loss of steel protection
- Depth = cover: no construction tolerance
- Strength slack 0.48 MPa; corner error 4.6 MPa
- Chloride (coastal) not modelled

Takeaway

- Use less clinker within constraints
- Baseline + modeled additive: 6–7% less carbon per m³
- Next: lab batches to confirm
- Questions?